Week 1 Internship Report

CityScape – Mini City Builder

Intern Name: Nathanieal Sandipurti

Project Title: CityScape – Mini City Builder

Week: 1

Duration: Day 1 – Day 6

Github Folder: https://github.com/Nathanieal06/CityScape

Objective

This week was about turning a rough idea into something I could actually build on. My goal was to lock down the concept, get Unity set up properly, gather and organize the assets I'd need, and get the core foundation of a city-building game in place — terrain, roads, grid-based building placement, and a basic UI.

Daily Work Log

Day 1 – Project Planning and Research

- Spent time exploring different game ideas that would work well for the internship timeline.
- Settled on CityScape – Mini City Builder as the project.
- Mapped out the scope and thought through the core gameplay mechanics I wanted to include.
- Looked at a few existing city-building games for inspiration and noted down the systems that seemed essential.

Outcome: Had a finalized concept and a rough development roadmap to work from.

Day 2 – Asset Collection

- Sourced a set of low-poly environment assets.
- Imported residential, industrial, commercial, utility, decoration, road, and terrain assets into Unity.
- Sorted everything into clearly named folders so the project wouldn't turn into a mess later.

Outcome: A usable asset library and a project structure that made sense.

Day 3 – Project Structure and Environment Setup

- Built the main gameplay scene.
- Blocked out the map using terrain, mountains, trees, and other environment pieces.
- Sketched out the overall city layout.
- Planned the architecture for the systems I'd need going forward: grid system, road placement, building placement, camera controller, UI structure, economy manager, and save/load system.

Outcome: The project structure was in place and ready for actual gameplay code.

Day 4 – Core Gameplay Implementation

- Got the grid-based placement system working.
- Built out road placement functionality.

- Added building placement with grid snapping.
- Implemented object rotation before placing objects down.

Outcome: Basic city construction was functional for the first time.

Day 5 – UI and Gameplay Features

- Built the building category panel.
- Added Residential, Industrial, Utility, Services, Decoration, and Road menus.
- Added a building information panel.
- Implemented camera controls.
- Added Build and Explore modes.

Outcome: A working gameplay interface came together.

Day 6 – Game Systems Integration and Testing

- Implemented save and load functionality.
- Added a pause menu.
- Put together a basic economy system.
- Added a minimap.
- Tested everything and fixed a handful of placement bugs.
- Cleaned up the scene layout and UI.

Outcome: The first playable prototype was complete.

Technologies Used

- Unity 6
- C#
- Unity UI System
- Git & GitHub
- Git LFS
- Low-Poly Asset Packs

Features Completed

- Low-poly environment setup
- Terrain and environment design
- Road construction system
- Grid-based building placement
- Building rotation
- Residential buildings
- Industrial buildings
- Utility buildings
- Service buildings
- Decoration system
- Build and Explore modes
- Pause menu
- Save/Load system

- Minimap
- Building information panel
- Basic economy UI

Challenges Faced

- Figuring out a project scope that was realistic for the internship's timeframe.
- Keeping a large number of imported assets organized.
- Designing a grid placement system that could actually scale.
- Wiring up multiple gameplay systems together without the project turning messy.

Solutions Implemented

- Trimmed the scope down to focus on the core city-building mechanics.
- Organized assets into categorized folders from the start.
- Used a modular architecture so systems could be built and tested independently.
- Built and tested each feature on its own before integrating it with the rest.

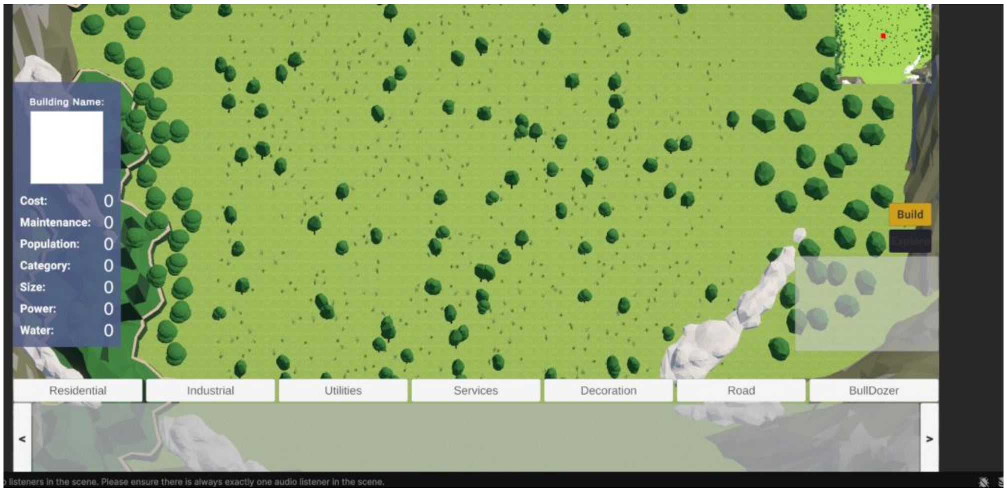
Conclusion

By the end of the first week, CityScape had gone from a bare idea to a working prototype. There's now a structured environment, grid-based building placement, road construction, categorized building menus, save/load functionality, and a basic UI in place. It's a solid base to build on for city simulation mechanics, resource management, and more advanced gameplay features in the coming weeks.

Screenshot Reference

A few screenshots taken along the way, showing how the project moved from an early prototype to a more complete build.

Day 1 — early road-building prototype, with placeholder UI buttons for roads, factories, shops, and houses.



Day 2 — terrain and environment setup, with the building info panel and category tabs taking shape.

Day 3 — structured map with , money system, and category menu in place.

Day 4–5 — building placement in action, with rotation controls and per-category costs visible.

Day 6 — the integrated prototype: a fully laid-out city with roads, zoned buildings, and a detailed minimap.

Expected Deliverables

For this submission, the following needs to be included:

- A Unity project folder containing a scene with at least one terrain.
- A few imported 3D models (free assets are acceptable).
- Proper asset organization in folders.
- A README file explaining the project structure and the overall approach, submitted in PDF format.

Key Steps

- Open the Project
- Click on Assets\ Scene\ SampleScene
- After Open that Scene Click Game Window Play Button
- Click Pause UI Button
- Then Click Load
- Map will be load which I made during my 1 Week of Unity Game Development Trainee Internship